

predicted values of AI adoption, which must be reflected in the standard errors of the OLS estimator. In our setting, AI adoption is constructed through a nonparametric matching procedure, which complicates the derivation of a closed-form or plug-in correction for standard errors. To address this issue of inference, for the main regressions with US-matched adoption rates, we report bootstrap standard errors, which is a valid procedure in this context. Specifically, we generate 500 bootstrap replications by resampling US firms with replacement within each size and sector stratum, while keeping the EU sample fixed. For each bootstrap replica, the matching is re-performed for every EU firm, and the same OLS specification as presented in Equation (1) is re-estimated. Standard errors are then computed from the bootstrap replications.⁹ In other specifications, we report standard errors clustered at the country-sector-wave level.

5 Results

Table 2 presents our main results. The first two columns use labor productivity (LP) as the dependent variable and the actual AI adoption rates of EU firms as the key explanatory variable of interest, without and with controls respectively. The results from column (1) suggest that, conditional on size, age and country-sector-wave fixed effects, adoption by EU firms raises labor productivity by about 16%, a very large number even when considering it is a one-off increase estimated from a pooled cross-section of firms over a period of five years. When controlling for time-varying firm-specific characteristics such as investment, profitability, leverage and size, the effect of adoption is significantly reduced in size but remains statistically significant. The estimates from column (2) suggest that more profitable and larger firms tend to have higher labor productivity, whereas more leveraged firms exhibit lower productivity.

The third and fourth columns present our key results, when using predicted AI adoption through the matching procedure using US firms described in Section 4. The estimates in column (3) do not include firm controls and suggest that the effect of instrumented AI adoption on labor productivity remains high, as in column (1). When controlling for firm characteristics, the effect of AI adoption on firm-level labor productivity is reduced but remains statistically significant. The estimate shows that AI adoption leads to an

⁹Table 2 reports statistical significance based on the normal approximation, although the results remain robust when using empirical p-values.

increase in labor productivity by 4% when instrumenting for AI adoption – a plausible figure that aligns with the more moderate effects found in the literature and discussed in Section 2. For the firm-specific controls, the results remain consistent with those in column (2). An exception is investment, which is now also statistically significant and positive (i.e. firms that invest more are also more productive).

Table 2: Effects of AI adoption on labor productivity

	(1)	(2)	(3)	(4)
	LP (log)	LP (log)	LP (log)	LP (log)
	No controls	With controls	No controls	With controls
AI adoption (actual)	0.162*** (0.011)	0.028*** (0.011)		
AI adoption (US-based)			0.156*** (0.03)	0.041*** (0.019)
Investment (lagged)		0.009* (0.005)		0.009*** (0.003)
Profitability (lagged)		0.590*** (0.038)		0.618*** (0.019)
Financial leverage (lagged)		-0.178*** (0.025)		-0.151*** (0.012)
Total assets (log, lagged)		0.386*** (0.006)		0.386*** (0.002)
Size FE	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes
Country-sector-wave FE	Yes	Yes	Yes	Yes
Observations	53730	35648	44844	30228
R-squared	0.373	0.541	0.380	0.551
Adjusted R-squared	0.352	0.519	0.355	0.526

Note: The dependent variable is labor productivity, calculated as the log of turnover per employee, and is derived from EIBIS. AI adoption (actual) is the EIBIS-reported AI implementation status. AI adoption (US-matched) is the AI implementation status derived from similar firms in the US. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level in columns (1) and (2), and bootstrapped in columns (3) and (4) are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.

Table 3: Effects of AI adoption on employment

	(1)	(2)	(3)	(4)
	Empl. (log)	Empl. (log)	Empl. (log)	Empl. (log)
	No controls	With controls	No controls	With controls
AI adoption (actual)	0.125*** (0.006)	0.079*** (0.006)		
AI adoption (US-based)			0.053*** (0.010)	-0.012 (0.010)
Investment (lag)		-0.003 (0.003)		-0.003 (0.003)
Profitability (lag)		0.153*** (0.019)		0.167*** (0.022)
Financial leverage (lag)		0.034** (0.014)		0.045*** (0.015)
Total assets (log, lag)		0.178*** (0.003)		0.189*** (0.003)
Size FE	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes
Country-sector-wave FE	Yes	Yes	Yes	Yes
Observations	55020	36259	45340	30446
R-squared	0.915	0.926	0.912	0.924
Adjusted R-squared	0.912	0.922	0.908	0.920

Note: The dependent variable is the log of number of employees and is derived from EIBIS. AI adoption (actual) is the EIBIS-reported AI implementation status. AI adoption (US-matched) is the AI implementation status derived from similar firms in the US. Investment is expressed as the annual change in total fixed assets. Profitability is the ratio of EBIT to total assets. Financial leverage is the ratio of loans and long-term debt to total assets. All control variables come from ORBIS and are lagged by 1 year. Standard errors clustered at the country-sector-wave level are reported in parentheses. *, **, *** correspond to 0.1, 0.05 and 0.01 significance levels, respectively.

We next examine whether the improvement in labor productivity stems from a reduction in employment. The results indicate that this is not the case. Table 3 reports the results, following a structure similar to Table 2, but with the logarithm of employment as the dependent variable. When using the actual series of AI adoption, the estimates suggest a positive effect on employment of 8% without controls (column (2)). In con-

trast, when instrumenting AI adoption using our PSM strategy and controlling for firm characteristics, we find no significant effect on employment (column (4)).

Taking the results from Tables 2 and 3 together begs the question whether the increase in labor productivity translates into benefits for workers. Table 4 addresses this issue by replicating our main regression with controls and all fixed effects, but using either the total wage bill or the average wage per employee, both at the firm level. The results show that AI-adopting firms display higher wages (overall and per employee) than non-adopting firms, after controlling for age, size and country-sector-wave fixed effects. The results suggest that productivity gains have so far benefited employees. It remains to be seen whether the patterns will hold in the medium to long term.

Our empirical analysis yields three key insights regarding AI adoption by non-financial firms in Europe. First, the instrumental variable estimates confirm a positive causal effect of AI adoption on labor productivity. AI adoption raises productivity by 4%, an effect that remains statistically significant even after controlling for firm-specific characteristics such as investment intensity, profitability, leverage and size.¹⁰

While direct comparisons with the macroeconomic literature are not straightforward – given that our results focus on labor productivity and reflect a one-off effect, whereas macro studies typically estimate long-run impacts on total factor productivity (TFP) – our findings appear more consistent with mid-range projections (e.g., [Bergeaud \(2024\)](#)’s 0.3 percentage points annual TFP gains or [Aghion and Bunel \(2024\)](#)’s 0.7 percentage points). The magnitude suggests that AI acts as a complementary input that enhances efficiency, though its aggregate impact is moderated by implementation frictions and skill gaps, contrasting with more optimistic projections of transformative productivity booms (e.g., [Baily, Brynjolfsson, and Korinek \(2023\)](#)’s 1.7 percentage points scenario). The increase in the coefficient from 2.8% (before matching) to 4.1% (after matching) indicates that endogenous adoption – whereby not all productive firms self-select into AI – understates perceived benefits. After isolating exogenous variation by benchmarking with the US adoption rates, the net effect remains positive, significant and economically plausible, reflecting incremental gains from task automation and cognitive augmentation.

¹⁰In the baseline specification, labor productivity is expressed as turnover per employee. However, the effect remains significant at 3.4% when use instead value added per employee (see Table B.3 in the Appendix).